7.1 Food biotechnology

Genetically modified food is synthesized using biotechnological tools. Modern Biotechnology is also called as genetic engineering, genetic modification or transgenic technology. Food bio technology is defined as ***"Application of technology to modify genes of animals, plants and micro organisms to create new species which have desired production, marketing, or nutrition related properties".*** In this technology, Nuclear DNA is modified through insertion of gene of interest (gene encoding desired trait). This modified DNA is called as recombinant DNA. When r**ecombinant DNA** expresses, it encodes desired product. This technology, when implemented enhance food qualities or yield is called as food bio technology

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7.1.1 Recent Food Technology

Developments in food technology have contributed greatly to the food supply and have changed our world. Some of these developments are:

- **Pharma food :** food or nutrient that claims medical or health benefits.
- **Anti oxidants:** It is capable of stabilizing or deactivating free radical before they attack cells.
- **Chemoprevention:** It uses one or several compounds to prevent, strip or reverse the development of cancer.
- **Designer food :** Processed food that are supplemented with food ingredients naturally rich in disease preventing substances.
- **Functional food:** any modified food or food ingredient that may provide a health benefit beyond the nutrients it contains.

7.1.2 Genetically modified foods

Genetically modified organisms (GMOs) can be defined as "Organisms (i.e. plants, animals or microorganisms) in which the genetic material (DNA) has been altered in a way that does not occur naturally by mating and/or natural recombination". The technology is often called "modern biotechnology" or "gene technology", sometimes also called as *"recombinant DNA technology" or "genetic engineering"*. It allows selected individual genes to be transferred from one organism into another, also between nonrelated species. Foods produced from or using GM organisms are often referred to as GM foods.

7.2 Nutraceuticals

`The term Nutraceuticals is a hybrid or contraction of nutrition and pharmaceuticals. Nutraceuticals are products derived from food sources that are purported to provide additional health benefits, in addition to the basic nutritional value found in foods. It is classified into two:

1. Dietary supplements
2. Functional foods

DO YOU KNOW...?

The word Nutraceuticals was coined by Dr. Stephen L. Defelice founder and chairman of the foundation for Innovation in Mountain side.

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